

Appendix C

Supplementary material for

ECDSA.Fail: Open Autoresearch for Optimizing Elliptic-Curve Point Addition in Shor's Algorithm

This document reproduces Appendix C of the paper. Section, figure, and table numbers are those of the main paper, so a reference such as Section 5.2.3 or Table 5 carries the same number in both documents. References to material outside Appendix C resolve in the main paper.

C Optimization Census of Accepted Competition Source Commits

This appendix lists every accepted source commit of the public competition (Section B), assigned to its primary transformation family or residual bucket—the same census summarized in Figure 10, here expanded to one row per commit. Official operating points—*Score* = average executed Toffoli $\times$ peak qubits (lower is better) and Δ , the score change versus the prior public frontier—are joined from the ecdsafail leaderboard; they are not in the git history itself. Of the 404 accepted “Accept submission” commits through 18 July, 400 have leaderboard score records: 397 match the **promoted** rows in the 2 July ledger snapshot used for the headline operating points, and three are later scored commits. The remaining four are the contest’s **DUMMY_TOFFOLIS** bootstrap commits, which went through the acceptance plumbing but were never scored and are marked (*unscored*). Accepted source commits and current ledger-status rows are therefore related archival views, not interchangeable counting units. Each row’s assignment is made by an audit of its actual `git show diff` (not its commit subject or lever name, which are frequently unreliable), matching Figure 10. Rows are grouped by family in descending order of frequency—with the two residual buckets (verifier search, neutral) and the unscored bootstrap group placed last—then by change-type (level 2 of Figure 10), and within a change-type by score (best first).

The largest recorded frontier delta belongs to the two-jump inversion base (cddd5df, under Windowing), whose Δ is nearly one billion; it is the landed **DIALOG_GCD** reimplementation of the modular inverse. This ledger lists all 404 accepted commits, including the four unscored bootstrap rows—placed in their own *Unscored bootstrap* group—that Figure 10 excludes; on the 400 scored source commits the ledger and the figure agree family by family.

Table 8: Competition optimization census: all 404 accepted source commits (#883–1294) through 18 July, classified by primary family, with official leaderboard score and frontier delta where available. 400 scored; 4 unscored bootstrap commits.

Commit	Contestant	Date	Score	Δ
Dead-Code and Redundancy Elimination (244)				
<i>comparator/width truncation (149)</i>				
75851e6	bket7	2026-06-19	1,658,362,156	−319,484
57eb64b	benhuang2025	2026-06-19	1,669,072,000	−3,061,328
1caf521	mpjunior92	2026-06-16	1,672,963,776	−1,569,792
7bab51f	nasqret	2026-06-14	1,677,861,900	−498,420
f50ae3b	gnuchev	2026-06-14	1,678,360,320	−269,100
6878ef8	gnuchev	2026-06-14	1,678,629,420	−319,410
cf99209	nasqret	2026-06-13	1,683,083,736	−526,864
186d9d7	rubenmarcus	2026-06-12	1,699,653,032	−767,774
6777223	rubenmarcus	2026-06-12	1,700,420,806	−627,298
0d1d1e7	mpjunior92	2026-06-12	1,701,048,104	−293,062
8cd8104	jackylee0424	2026-06-11	1,702,054,020	−1,013,749
5a783e4	nasqret	2026-06-11	1,704,784,725	−270,945
f4404de	jackylee0424	2026-06-11	1,713,097,160	−26,840
1a5e620	BitWonka	2026-06-10	1,716,174,108	−26,999,973
00fb66d	jieyilong	2026-06-10	1,743,174,081	−623,931
707c1d9	jieyilong	2026-06-10	1,765,489,770	−632,220
8e0bf4f	BitWonka	2026-06-10	1,766,121,990	−2,044,435
ab5469b	jackylee0424	2026-06-10	1,774,726,350	−20,560
980ed3a	jackylee0424	2026-06-09	1,785,547,335	−1,286,285
2dcf00d	jackylee0424	2026-06-08	1,820,641,701	−28,534
f3820d0	SamrendraS	2026-06-08	1,823,175,078	−1,406,122
da55bbb	SamrendraS	2026-06-08	1,824,581,200	−694,200
73f4f48	gnuchev	2026-06-08	1,891,349,200	−1,475,700
a5a590a	gnuchev	2026-06-08	1,892,824,900	−109,934
94927be	jackylee0424	2026-06-07	1,893,823,629	−3,142,197
013f1f7	zuiris	2026-06-07	1,899,562,014	−1,666,560
99b9cff	zuiris	2026-06-07	1,901,228,574	−4,775,252
fc84586	jackylee0424	2026-06-07	1,911,704,516	−676,508
8925623	BitWonka	2026-06-07	1,939,751,375	−188,208
779fc2b	jackylee0424	2026-06-06	1,944,384,690	−677,026
ace6347	teddyjfpender	2026-06-06	1,945,061,716	−661,342
e0767c7	jackylee0424	2026-06-06	1,945,723,058	−677,026
3033d01	jackylee0424	2026-06-06	1,967,891,695	−675,444
88079e6	10d9e	2026-06-06	1,969,242,583	−675,444
8e13587	runeape-sats	2026-06-06	1,969,918,027	−141,372
ebb44e	zuiris	2026-06-06	1,977,629,346	−827,329
7f656bd	zuiris	2026-06-06	1,978,786,502	−4,018,630
1121d80	welttowelt	2026-06-06	1,982,805,132	−1,361,360
cee4044	welttowelt	2026-06-06	1,984,166,492	−1,511,895

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Commit	Contestant	Date	Score	Δ
c80fc1c	newjordan	2026-06-06	1,990,208,636	-1,123,928
974689f	newjordan	2026-06-06	1,991,332,564	-680,134
373fb4d	newjordan	2026-06-06	1,992,012,698	-680,134
9906ec0	newjordan	2026-06-06	1,992,692,832	-443,794
2f1f175	newjordan	2026-06-06	1,993,136,626	-680,134
196d830	newjordan	2026-06-06	2,013,389,651	-1,397,032
22624a0	Bit Wonka	2026-06-06	2,014,786,683	-1,150,188
587b6d8	anshu4321	2026-06-06	2,015,936,871	-680,134
4ef0046	Bit Wonka	2026-06-05	2,016,617,005	-212,706
02049b8	10d9e	2026-06-05	2,016,829,711	-1,150,188
91061af	newjordan	2026-06-05	2,020,184,357	-1,838,686
5ad4427	newjordan	2026-06-05	2,022,023,043	-1,688,320
58ecf0a	10d9e	2026-06-05	2,025,245,640	-633,600
351a0d0	10d9e	2026-06-05	2,025,879,240	-1,151,040
83da96f	10d9e	2026-06-05	2,027,030,280	-3,429,360
ec0fa98	10d9e	2026-06-05	2,030,459,640	-2,808,960
e4497a6	robertkodra	2026-06-05	2,036,687,400	-1,156,320
db60686	robertkodra	2026-06-05	2,037,843,720	-2,595,120
bbf7b3a	AnalyticETH	2026-06-05	2,046,711,480	-464,640
8c28287	zuiris	2026-06-05	2,047,176,120	-1,277,760
8a9d274	factory-sagar	2026-06-05	2,048,453,880	-781,440
2f24e2f	robertkodra	2026-06-05	2,049,235,320	-1,156,320
6abbe60	robertkodra	2026-06-05	2,050,391,640	-364,320
cf5e390	robertkodra	2026-06-05	2,050,755,960	-1,140,480
c3cbcce	newjordan	2026-06-05	2,051,896,440	-728,640
5ad23b1	zuiris	2026-06-05	2,052,625,080	-1,029,210
ca7fd40	mpjunior92	2026-06-05	2,055,555,480	-4,812,720
229aeeed	10d9e	2026-06-05	2,060,368,200	-498,960
a92d098	newjordan	2026-06-05	2,064,288,600	-192,720
94e9418	newjordan	2026-06-05	2,064,481,320	-1,869,120
ae3b97b	10d9e	2026-06-05	2,091,370,926	-530,688
05d68e5	10d9e	2026-06-05	2,091,901,614	-751,808
fa5347c	factory-sagar	2026-06-05	2,110,268,810	-1,442,820
bc858e8	10d9e	2026-06-05	2,111,711,630	-720,020
fc27501	alexander-sei	2026-06-05	2,112,431,650	-13,577,520
f0e2891	robertkodra	2026-06-05	2,126,009,170	-1,237,100
b84d15c	robertkodra	2026-06-05	2,127,246,270	-1,334,400
af0f90a	10d9e	2026-06-05	2,128,580,670	-720,020
dda7bfd	robertkodra	2026-06-05	2,129,300,690	-811,760
98416e4	robertkodra	2026-06-05	2,130,112,450	-261,320
4527ae1	PhantasticUniverse	2026-06-05	2,130,373,770	-1,073,080
1991d1c	newjordan	2026-06-05	2,132,814,610	-17,525,120
7c82c3c	newjordan	2026-06-05	2,150,339,730	-4,692,640
247db26	johnbxx	2026-06-05	2,156,477,970	-39,476,000
5b3887b	newjordan	2026-06-05	2,195,953,970	-25,242,400
c6cc74e	robertkodra	2026-06-05	2,222,564,130	-5,604,480

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Commit	Contestant	Date	Score	Δ
01260e8	newjordan	2026-06-05	2,228,168,610	-29,556,960
9d30eb7	johnbxx	2026-06-05	2,257,725,570	-8,651,360
d3e35ae	mpjunior92	2026-06-03	2,405,410,905	-1,701,880
cbe70bc	mpjunior92	2026-06-03	2,407,112,785	-2,829,240
353ac6d	mpjunior92	2026-06-03	2,409,942,025	-693,760
da42ce2	harrigan	2026-06-03	2,410,635,785	-2,905,120
5b5a5d4	teddyjfpender	2026-06-03	2,430,923,007	-3,025,184
b27f99c	aburan28	2026-06-03	2,455,281,894	-6,619,344
bdfc7df	aburan28	2026-06-03	2,471,488,962	-5,019,000
e75ea5c	10d9e	2026-06-03	2,476,507,962	-1,542,984
2bd707e	aburan28	2026-06-03	2,478,050,946	-1,497,096
b343178	BitWonka	2026-06-03	2,479,548,042	-648,168
5d2f6cf	BitWonka	2026-06-03	2,480,196,210	-1,132,860
238d366	BitWonka	2026-06-03	2,481,329,070	-1,525,776
34a514a	twangodev	2026-06-03	2,484,807,954	-1,135,728
5a2d910	twangodev	2026-06-03	2,485,943,682	-7,995,984
113f346	twangodev	2026-06-03	2,498,876,289	-6,632,028
144e07f	phileigenlabs	2026-06-03	2,505,508,317	-473,304
d2b4bcb	Epistetechnician	2026-06-02	2,510,523,510	-1,469,136
14b84f1	10d9e	2026-06-02	2,533,577,850	-1,489,456
47f30d4	josusanmartin	2026-06-02	2,592,421,194	-1,467,984
1207861	pldallairedemers	2026-06-02	2,637,203,021	-1,491,784
ecbf87b	mpjunior92	2026-06-02	2,638,694,805	-2,507,200
006e38b	solimander	2026-06-02	2,641,202,005	-1,250,466
7408de1	prasiddhnaik	2026-06-02	2,736,895,656	-58,127
6e7ca2c	mpjunior92	2026-06-02	2,736,953,783	-1,822,360
14fc786	mpjunior92	2026-06-02	2,738,776,143	-2,934,628
bb27ddd	anupsv	2026-06-02	2,769,217,908	-14,632,176
4b68e94	mpjunior92	2026-06-02	2,822,498,802	-1,215,768
412312f	mpjunior92	2026-06-02	2,823,714,570	-6,656,160
8316db1	mpjunior92	2026-06-02	2,830,370,730	-1,215,768
1fd942d	Gajesh2007	2026-06-02	2,831,586,498	-1,086,720
eda50f2	Epistetechnician	2026-06-02	2,832,673,218	-869,376
b694e22	Gajesh2007	2026-06-02	2,833,542,594	-2,859,432
dd103cd	M3kko	2026-06-02	2,873,320,661	-9,151,501
ad02935	Gajesh2007	2026-06-02	2,882,472,162	-9,659,922
27530d2	pldallairedemers	2026-06-02	2,892,132,084	-1,405,944
1846bf7	megabyte0x	2026-06-02	3,270,229,140	-4,265,376
3c30191	jtrose24	2026-06-02	3,315,144,636	-49,839,696
6447611	owizdom	2026-06-02	3,538,424,844	-236,823,456
8cb350c	Gajesh2007	2026-06-02	4,068,676,284	-27,833,616
564c4c3	owizdom	2026-06-02	4,096,509,900	-34,503,360
00e3a20	anupsv	2026-06-02	4,131,013,260	-17,007,168
541bd85	josusanmartin	2026-06-02	4,148,020,428	-8,422,080
700d5a7	0xpg	2026-06-02	5,148,949,806	-8,808,800
6e71096	Gajesh2007	2026-06-02	5,561,829,149	-849,712

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Commit	Contestant	Date	Score	Δ
85ee8e7	mpjunior92	2026-06-01	5,574,826,510	-10,256,578
5136c0a	mpjunior92	2026-06-01	5,597,330,024	-25,066,504
49b7969	anupsv	2026-06-01	5,622,396,528	-10,071,858
141b79a	Gajesh2007	2026-06-01	5,935,088,235	-40,093,476
9fa5a56	anupsv	2026-06-01	5,976,793,393	-51,347,542
c7e1ae5	saucegodbased	2026-06-01	6,028,140,935	-7,157,900
a25248f	Gajesh2007	2026-06-01	6,035,298,835	-4,811,956
404d630	Gajesh2007	2026-06-01	6,040,110,791	-413,504,956
0c25aca	Gajesh2007	2026-06-01	6,453,615,747	-8,280,074
bf19c9e	Gajesh2007	2026-06-01	6,461,895,821	-27,320,088
6eba4e3	Gajesh2007	2026-06-01	6,489,215,909	-3,204,892
e739b53	anupsv	2026-06-01	6,540,623,485	-10,196,544
179ab10	anupsv	2026-06-01	6,550,820,029	-13,535,358
fc8b9e6	Gajesh2007	2026-06-01	6,616,811,249	-10,113,420
13ac8da	Gajesh2007	2026-06-01	6,764,917,436	-100,441,500
b86e789	Gajesh2007	2026-06-01	6,865,358,936	-96,285,300
6da9882	Gajesh2007	2026-06-01	6,964,659,240	-110,103,840
36e8578	Gajesh2007	2026-06-01	7,077,100,800	-677,010,180
<i>iteration/threshold tuning (37)</i>				
f861f87	jackylee0424	2026-06-20	1,597,837,965	-1,159
5f3a0e3	bket7	2026-06-20	1,598,512,503	-207,461
0f0c1c2	bket7	2026-06-20	1,598,719,964	-1,335,168
0fa5c6f	jieyilong	2026-06-13	1,683,610,600	-1,371,863
f6f9536	BitWonka	2026-06-10	1,768,166,425	-6,559,925
1a7895c	welttowelt	2026-06-07	1,907,351,618	-4,352,898
2c3b795	zuiris	2026-06-06	1,940,606,153	-3,778,537
f5f0f61	jieyilong	2026-06-06	1,949,378,508	-3,776,465
a6a92a4	jieyilong	2026-06-06	1,953,154,973	-3,724,105
f37a0bc	jackylee0424	2026-06-06	1,956,879,078	-3,734,577
76fdaf2	newjordan	2026-06-06	1,986,336,599	-3,872,037
c64c03f	PhantasticUniverse	2026-06-05	2,060,867,160	-3,421,440
fec6d25	10d9e	2026-06-05	2,092,653,422	-17,615,388
8ef64e9	stevenhao	2026-06-04	2,381,382,450	-18,066,455
39134ec	stevenhao	2026-06-04	2,399,448,905	-2,981,000
c3c3f8d	PhantasticUniverse	2026-06-04	2,402,429,905	-2,981,000
a392fd4	aburan28	2026-06-03	2,439,135,048	-3,078,384
311d6c3	aburan28	2026-06-03	2,445,362,892	-3,038,784
339fe8d	aburan28	2026-06-03	2,482,854,846	-1,953,108
f42ce5c	Sam-Scolari	2026-06-03	2,497,479,574	-1,396,715
fd8e712	pdlallairedemers	2026-06-02	2,535,067,306	-4,459,572
7c29f79	mmurrs	2026-06-02	2,574,991,500	-3,084,000
801b73e	0xpg	2026-06-02	5,157,758,606	-904,904
13a7a53	antojoseph	2026-06-02	5,182,912,575	-421,200
17ae622	Epistetechnician	2026-06-02	5,183,333,775	-1,684,800
0e726ac	mpjunior92	2026-06-02	5,547,081,566	-1,339,220
8eb6b5e	jtrose24	2026-06-02	5,562,678,861	-12,147,649

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Commit	Contestant	Date	Score	Δ
297f5e5	jtrose24	2026-06-01	5,653,140,863	-33,727,563
330186c	anupsv	2026-06-01	5,975,181,711	-1,611,682
ac52c2e	Gajesh2007	2026-06-01	6,496,336,865	-11,720,484
f910ba3	Gajesh2007	2026-06-01	7,756,453,320	-1,173,480
a428a8c	Gajesh2007	2026-06-01	7,757,626,800	-8,036,490
77b87f1	bxue-l2	2026-06-01	8,372,879,130	-25,902,869
a3ee589	philuponcrypto	2026-05-31	8,405,420,100	-40,029,990
1c400f2	chainyoda	2026-05-31	10,318,536,120	-3,910,352
a444668	unknown	2026-05-31	10,397,877,812	-690,540
9bd0748	unknown	2026-05-31	10,398,568,352	-173,734,448
<i>fold/carry truncation (37)</i>				
f6ce76d	BitWonka	2026-06-20	1,597,839,124	-229,482
3df690f	gopikannappan	2026-06-20	1,607,030,040	-1,870,580
31421df	jieyilong	2026-06-20	1,616,813,772	-1,877,220
f265bb1	jackylee0424	2026-06-19	1,646,703,035	-9,320
550595a	jackylee0424	2026-06-19	1,649,234,708	-1,166
ed2715f	gnuchev	2026-06-19	1,649,235,874	-461,736
764afda	gnuchev	2026-06-19	1,651,099,142	-39,644
916c0e3	DavetheSlayer	2026-06-19	1,651,138,786	-37,312
bc2334a	BitWonka	2026-06-19	1,651,176,098	-5,923,280
fd89e69	jieyilong	2026-06-19	1,658,681,640	-26,818
a9ac507	jieyilong	2026-06-19	1,658,708,458	-26,818
c9c03de	jackylee0424	2026-06-19	1,658,735,276	-4,664
98e1322	zuiris	2026-06-11	1,709,962,980	-3,134,180
c8f4536	BitWonka	2026-06-07	1,939,939,583	-666,570
11ceae2	welttowelt	2026-06-06	1,960,613,655	-7,278,040
004ffc0	zuiris	2026-06-06	1,970,059,399	-1,350,888
69bd3d7	Gajesh2007	2026-06-06	1,971,410,287	-5,931,008
21fbfc7	robertkodra	2026-06-05	2,033,268,600	-3,418,800
66c6dac	mpjunior92	2026-06-05	2,054,166,840	-1,388,640
438ef00	mpjunior92	2026-06-05	2,089,123,794	-2,247,132
5174a69	PhantasticUniverse	2026-06-05	2,131,446,850	-1,367,760
657ae28	PhantasticUniverse	2026-06-05	2,221,196,370	-1,367,760
cc5584e	PhantasticUniverse	2026-06-05	2,379,298,050	-2,084,400
40574d3	mpjunior92	2026-06-02	2,616,750,555	-3,693,942
7787f81	mpjunior92	2026-06-02	2,620,444,497	-10,554,777
f8a36ba	Gajesh2007	2026-06-02	2,893,538,028	-227,470,872
115ab0e	xadenryan	2026-06-02	5,166,788,002	-32,096
61bdeb0	antojoseph	2026-06-02	5,180,304,536	-48,576
8d83e8e	0xpg	2026-06-02	5,182,432,650	-479,925
3ab1c72	anupsv	2026-06-02	5,548,420,786	-1,108,320
3f534f2	Gajesh2007	2026-06-02	5,549,529,106	-12,300,043
1b913c5	anupsv	2026-06-01	5,585,083,088	-12,246,936
ba05d3a	Gajesh2007	2026-06-01	5,686,868,426	-195,285,984
d6551e8	solimander	2026-06-01	5,882,154,410	-52,933,825
a97b4e9	Gajesh2007	2026-06-01	6,508,057,349	-18,430,438

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Table 8, continued

Commit	Contestant	Date	Score	Δ
2abe7f4	Gajesh2007	2026-06-01	6,564,355,387	-52,455,862
1e18b4d	Gajesh2007	2026-06-01	6,626,924,669	-137,992,767
<i>dead-gate/self-inverse removal (21)</i>				
9eea46c	jackylee0424	2026-07-18	1,519,161,984	-8,064
a505e6d	mochimodev	2026-07-11	1,519,170,048	-2,348,928
6dafa07	BitWonka	2026-06-26	1,576,955,794	-894,728
da51a48	jieyilong	2026-06-23	1,577,865,511	-97,609
985eaec	BitWonka	2026-06-23	1,577,963,120	-1,086,640
27d4627	BitWonka	2026-06-22	1,579,049,760	-321,368
6c0769f	BitWonka	2026-06-22	1,579,371,128	-1,945,292
20b9a1d	mpjunior92	2026-06-21	1,581,316,420	-1,439,478
4a90d04	zuiris	2026-06-21	1,582,755,898	-13,950,606
9af02f7	gnuchev	2026-06-21	1,597,055,640	-282,796
9d524b7	BitWonka	2026-06-20	1,597,538,943	-150,787
6ba606a	jieyilong	2026-06-20	1,597,689,730	-148,235
a47dc6e	PhantasticUniverse	2026-06-19	1,644,083,652	-1,186,116
ba3efc1	BitWonka	2026-06-18	1,672,133,328	-830,448
7822c37	BitWonka	2026-06-09	1,774,746,910	-1,322,265
b8ce940	BitWonka	2026-06-09	1,776,069,175	-651,495
b55ede3	welttowelt	2026-06-06	1,968,567,139	-226,336
975b73f	josusanmartin	2026-06-02	2,590,854,522	-1,515,843
908521d	anupsv	2026-06-02	5,121,484,498	-27,465,308
c8e508f	Gajesh2007	2026-06-01	8,367,457,560	-5,421,570
a16788e	unknown	2026-05-30	10,579,872,520	-118,867,397
Register and Scratch Allocation (50)				
<i>ancilla-lifetime / venting (30)</i>				
d11bdbb	BitWonka	2026-06-20	1,600,244,049	-6,765,039
fed64cf	nasqret	2026-06-20	1,608,900,620	-5,159,212
b310de9	BitWonka	2026-06-19	1,642,623,526	-924,686
f8d23a9	BitWonka	2026-06-19	1,645,269,768	-1,030,177
cea9f5f	BitWonka	2026-06-19	1,646,712,355	-2,059,451
cf3c6f1	BitWonka	2026-06-12	1,701,341,166	-712,854
9191f81	zuiris	2026-06-11	1,708,341,222	-1,621,758
572bba4	zuiris	2026-06-10	1,743,798,012	-3,488,606
da398df	BitWonka	2026-06-08	1,786,833,620	-11,871,920
89abe1b	zuiris	2026-06-06	1,946,400,084	-2,978,424
33c072d	jieyilong	2026-06-06	1,968,793,475	-449,108
a5e8bdf	runeape-sats	2026-06-06	1,977,341,295	-288,051
571e67c	runeape-sats	2026-06-06	1,978,456,675	-329,827
06b07ee	teddyjfpender	2026-06-05	2,066,350,440	-22,773,354
1eae277	newjordan	2026-06-05	2,155,032,370	-1,445,600
250774e	10d9e	2026-06-02	2,578,075,500	-12,779,022
46c3e3a	bxue-l2	2026-06-02	2,593,889,178	-1,682,159
7021abe	anupsv	2026-06-02	2,614,737,018	-544,601
287595e	10d9e	2026-06-02	2,615,519,241	-1,231,314

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Table 8, continued

Commit	Contestant	Date	Score	Δ
550895c	solimander	2026-06-02	2,647,454,335	−84,484,816
36dcf6b	pdlallairedemers	2026-06-02	2,744,218,087	−1,746,797
68e3653	saucegodbased	2026-06-02	2,783,850,084	−38,648,718
0435f04	nandy-technologies	2026-06-02	5,160,257,102	−5,760,596
ccd7530	Gajesh2007	2026-06-02	5,166,820,098	−13,484,438
9899c06	antojoseph	2026-06-02	5,180,353,112	−2,079,538
233a44c	Gajesh2007	2026-06-02	5,185,018,575	−362,062,991
0e32587	Gajesh2007	2026-06-01	6,961,644,236	−3,015,004
74a1a8b	cryptogakusei	2026-05-31	9,603,905,752	−433,280
b827c9e	unknown	2026-05-31	10,395,860,352	−2,017,460
bc9728a	unknown	2026-05-30	10,572,584,432	−7,288,088
<i>scratch/ancilla reuse (17)</i>				
461a4a3	BitWonka	2026-06-11	1,703,067,769	−1,716,956
420e0c2	BitWonka	2026-06-11	1,705,055,670	−2,814,186
997dbd6	BitWonka	2026-06-11	1,713,124,000	−3,050,108
8983da8	SamrendraS	2026-06-08	1,822,107,738	−1,067,340
ea8ce7b	jieyilong	2026-06-07	1,916,455,744	−1,488,336
3409fbd	zuiris	2026-06-07	1,920,798,194	−1,491,661
13a01b8	nikhiljha	2026-06-03	2,413,540,905	−17,382,102
e0cfe2b	nikhiljha	2026-06-03	2,433,948,191	−3,091,060
24a4076	nikhiljha	2026-06-03	2,437,039,251	−2,095,797
3166bbd	nikhiljha	2026-06-03	2,442,213,432	−3,149,460
dd188f8	nikhiljha	2026-06-03	2,448,401,676	−6,880,218
1d2cbc9	mpjunior92	2026-06-02	2,505,981,621	−2,913,693
6189d5f	alexander-sei	2026-06-02	2,508,895,314	−1,628,196
8ab5f90	pdlallairedemers	2026-06-02	2,511,992,646	−21,585,204
92a247e	mpjunior92	2026-06-02	2,539,526,878	−35,464,622
a8688f8	josusanmartin	2026-06-02	5,166,017,698	−770,304
666b591	Gajesh2007	2026-05-31	10,347,216,548	−48,643,804
<i>in-place mutation (3)</i>				
fe29b53	josusanmartin	2026-06-02	2,615,281,619	−237,622
b113c6f	mpjunior92	2026-06-02	2,630,999,274	−6,203,747
d19dbb5	Gajesh2007	2026-05-31	8,445,450,090	−1,145,844,506
Uncompute Strategy (22)				
<i>measurement uncompute (22)</i>				
d44cad3	BitWonka	2026-06-26	1,571,592,960	−5,362,834
5c34dd1	BitWonka	2026-06-20	1,598,068,606	−443,897
d636d62	nasqret	2026-06-11	1,707,869,856	−471,366
08c5068	BitWonka	2026-06-10	1,747,286,618	−12,457,386
b7515d5	zuiris	2026-06-09	1,780,286,545	−3,442,515
c27e8a5	zuiris	2026-06-09	1,783,729,060	−1,818,275
98dd2ad	BitWonka	2026-06-08	1,798,705,540	−21,936,161
bfd3fa6	BitWonka	2026-06-08	1,825,275,400	−66,073,800
e321f72	BitWonka	2026-06-07	1,892,934,834	−888,795
bde4caa	zuiris	2026-06-07	1,896,965,826	−2,596,188

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Table 8, continued

Commit	Contestant	Date	Score	Δ
a88c9a1	welttowelt	2026-06-07	1,906,003,826	-1,347,792
94535d1	welttowelt	2026-06-07	1,917,944,080	-2,854,114
fb83e6c	nikhiljha	2026-06-03	2,461,901,238	-9,587,724
985ae0f	josusanmartin	2026-06-02	2,841,822,042	-26,135,195
0aa33ed	Gajesh2007	2026-06-02	2,871,482,706	-1,837,955
acf2bdb	saucegodbased	2026-06-02	3,121,008,900	-40,650,120
4b9cfbb	saucegodbased	2026-06-02	3,274,494,516	-40,650,120
c813ee5	saucegodbased	2026-06-02	3,364,984,332	-173,440,512
7e660f9	saucegodbased	2026-06-02	3,775,248,300	-293,427,984
ce60f0e	Gajesh2007	2026-06-01	7,754,110,980	-2,342,340
81ff75b	Gajesh2007	2026-05-31	10,322,446,472	-21,997,084
e33b400	cryptogakusei	2026-05-31	10,344,443,556	-2,772,992
Constant Propagation and Classical Specialization (19)				
<i>host-derived / classical specialization (19)</i>				
cde752d	BitWonka	2026-06-21	1,596,706,504	-349,136
f8e215b	gopikannappan	2026-06-20	1,597,382,478	-156,465
662e267	bket7	2026-06-20	1,600,055,132	-188,917
7b7bd12	BitWonka	2026-06-20	1,614,062,156	-70,882
d2643bc	nasqret	2026-06-19	1,618,690,992	-377,136
fa7126d	gopikannappan	2026-06-19	1,643,656,464	-427,188
9798bc9	gopikannappan	2026-06-19	1,646,299,945	-356,490
b02b354	gopikannappan	2026-06-19	1,648,771,806	-462,902
497cc20	gopikannappan	2026-06-19	1,649,697,610	-1,378,212
c69b6c1	zuiris	2026-06-05	2,023,711,363	-1,534,277
8117d98	mpjunior92	2026-06-02	2,642,452,471	-5,001,864
17aad9a	mpjunior92	2026-06-02	2,731,939,151	-4,956,505
335ac78	solimander	2026-06-02	2,741,710,771	-2,507,316
075c0c6	ptrdsh	2026-06-02	2,745,964,884	-5,017,824
9805d0c	Epistetechnician	2026-06-02	2,836,402,026	-5,420,016
0261289	Epistetechnician	2026-06-02	2,867,957,237	-3,525,469
b60eec2	saucegodbased	2026-06-01	8,398,781,999	-6,638,101
3bbccd3	mmurrs	2026-05-31	9,591,294,596	-16,248
8631b04	unknown	2026-05-30	10,572,302,800	-281,632
Strength Reduction (14)				
<i>fast-adder reroute (9)</i>				
2a87f33	Gajesh2007	2026-06-09	1,776,720,670	-2,983,770
82f7a95	newjordan	2026-06-07	1,912,381,024	-4,074,720
639f12c	runeape-sats	2026-06-06	1,985,678,387	-658,212
df83d16	bxue-l2	2026-06-02	2,595,571,337	-19,165,681
53f0bdb	anupsv	2026-06-01	5,632,468,386	-20,672,477
d4959a1	Gajesh2007	2026-06-01	6,492,420,801	-3,916,064
0be8d11	Gajesh2007	2026-06-01	6,526,487,787	-14,135,698
04b3a76	mmurrs	2026-05-31	9,591,310,844	-12,594,908
168048c	unknown	2026-05-30	10,698,739,917	-5,834,478
<i>solinas / NAF fold (5)</i>				

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Table 8, continued

Commit	Contestant	Date	Score	Δ
aacf05c	gnuchev	2026-06-20	1,614,133,038	-1,831,866
e25c7d8	0xLucqs	2026-06-20	1,615,964,904	-848,868
4ea8b74	alexander-sei	2026-06-19	1,641,531,000	-1,092,526
ed94ad2	SamrendraS	2026-06-09	1,779,704,440	-582,105
6bf73fe	BitWonka	2026-06-07	1,922,289,855	-17,461,520
Fusion (12)				
<i>gate/op fusion (7)</i>				
12c7337	PhantasticUniverse	2026-06-19	1,643,554,032	-102,432
b1dec1e	nasqret	2026-06-19	1,657,099,378	-1,253,450
84d5b0e	rubenmarcus	2026-06-15	1,676,067,120	-1,794,780
8a47cf5	johnbxx	2026-06-06	1,993,816,760	-19,572,891
e4edca0	nikhiljha	2026-06-03	2,493,939,666	-3,539,908
9b6b9d6	Epistetechnician	2026-06-02	3,161,659,020	-108,570,120
f9fae31	Gajesh2007	2026-06-01	7,074,763,080	-2,337,720
<i>boundary cswap-merge (5)</i>				
422f21d	mochimodev	2026-07-07	1,521,518,976	-50,073,984
cef001a	owizdom	2026-06-02	5,120,690,122	-794,376
cd34b79	antojoseph	2026-06-02	5,158,663,510	-1,593,592
36a2989	Gajesh2007	2026-06-01	7,765,663,290	-601,794,270
9bcc27d	Gajesh2007	2026-05-31	9,604,339,032	-714,197,088
Windowing (8)				
<i>windowed carry/shift blocks (5)</i>				
7689eac	welttowelt	2026-06-10	1,759,744,004	-5,745,766
2eab923	PhantasticUniverse	2026-06-05	2,040,438,840	-6,272,640
a91b626	robertkodra	2026-06-05	2,053,654,290	-512,550
a772a45	robertkodra	2026-06-05	2,266,376,930	-5,077,764
7297f4f	fortunexbt	2026-06-02	2,592,370,365	-50,829
<i>K2 transcript pair-compressor (2)</i>				
9a81017	anshu4321	2026-06-05	2,017,979,899	-2,204,458
0780f67	johnbxx	2026-06-05	2,271,454,694	-107,843,356
<i>two-jump inversion base (1)</i>				
cdd5df	pdlallairedemers	2026-06-02	4,156,442,508	-964,247,614
Codec / Encoding (7)				
<i>trailmix / K5 codec (7)</i>				
ab1b2d6	PhantasticUniverse	2026-06-19	1,658,770,256	-1,389,940
bdb1d22	tob-joe	2026-06-19	1,660,163,697	-8,908,303
35ceb01	BitWonka	2026-06-15	1,674,533,568	-1,533,552
674d0d8	nasqret	2026-06-14	1,678,948,830	-2,076,765
cf310ec	nasqret	2026-06-13	1,681,025,595	-2,058,141
bb579bb	BitWonka	2026-06-13	1,684,982,463	-12,415,650
6953d1b	BitWonka	2026-06-12	1,697,398,113	-2,254,919
Algorithmic Substitution (2)				
<i>multiplier/squarer swap (2)</i>				
28fe2f2	alexander-sei	2026-06-19	1,619,139,132	-22,382,556

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Table 8, continued

Commit	Contestant	Date	Score	Δ
b6d6f41	mpjunior92	2026-06-02	2,750,982,708	−18,235,200
Verifier Search (19)				
<i>nonce/island reroll (19)</i>				
4cd1b2f	welttowelt	2026-06-25	1,577,850,522	−3,459
9b09c7e	welttowelt	2026-06-25	1,577,853,981	−3,459
d17af14	welttowelt	2026-06-25	1,577,857,440	−8,071
c07bf74	welttowelt	2026-06-21	1,597,338,436	−40,565
85ee6af	newjordan	2026-06-21	1,597,379,001	−3,477
42b56c6	welttowelt	2026-06-20	1,607,009,088	−20,952
94d44be	welttowelt	2026-06-20	1,614,059,832	−2,324
883f71a	nasqret	2026-06-19	1,619,068,128	−27,936
533ef88	gopikannappan	2026-06-19	1,619,096,064	−12,804
41df0b5	gopikannappan	2026-06-19	1,619,108,868	−17,460
3b05e0a	jackylee0424	2026-06-19	1,619,126,328	−12,804
eb089b1	gopikannappan	2026-06-19	1,641,521,688	−9,312
6a9db88	jackylee0424	2026-06-19	1,643,548,212	−5,820
fdb103e	jackylee0424	2026-06-19	1,646,656,435	−46,600
01560c7	gnuchev	2026-06-19	1,651,075,822	−23,320
70bf11a	rubenmarcus	2026-06-19	1,658,352,828	−9,328
5407215	jackylee0424	2026-06-19	1,658,739,940	−30,316
615ce9a	jackylee0424	2026-06-19	1,660,160,196	−3,501
a4db9a5	SamrendraS	2026-06-08	1,820,670,235	−1,437,503
Neutral / Mechanical (3)				
<i>DUMMY_TOFFOLIS padding change (3)</i>				
f43a73e	unknown	2026-05-30	10,704,574,395	−47,186,700
ed34c00	unknown	2026-05-30	10,751,761,095	−1,683,300
6f7c159	unknown	2026-05-30	10,753,444,395	0
Unscored bootstrap (DUMMY_TOFFOLIS) (4)				
15a06af	unknown	2026-05-29	(unscored)	—
971d801	unknown	2026-05-30	(unscored)	—
c55911f	unknown	2026-05-29	(unscored)	—
ca1bf33	unknown	2026-05-30	(unscored)	—

References